

CAAF: An NDN-Based Cache-Aware Adaptive Forwarding Strategy for Reliable Content Delivery in VANETs

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Abstract—The high mobility in Vehicular Ad-hoc Networks (VANETs) significantly affects the reliability of data transmission. To solve this problem, Named Data Networking (NDN)-based VANETs are proposed, utilizing in-network caching and named-based forwarding to overcome the dual challenges of mobility and connectivity. Although in-network caching enhances content availability, a strategy that accurately locates and efficiently utilizes the cached content in VANETs with highly dynamic environments is still lacking. In this paper, we propose a novel NDN-based cache-aware adaptive forwarding (CAAF) strategy for VANETs. CAAF proactively predicts content locations and ensures reliable content retrieval by adaptively selecting forwarding nodes that prioritize fast delivery and stable transmission. Specifically, we design a content information table for each vehicle to record information about the Interest packets it receives. Furthermore, these tables are updated periodically across all vehicles and a prediction model is used to predict real-time in-network caching during the update interval. Subsequently, we execute a filter mechanism to sieve candidate forwarding vehicles that satisfy both the accessibility and stability requirements. These candidates are then evaluated using a multi-attribute decision-making method across diverse parameters to determine the optimal forwarding node. Our extensive simulation results demonstrate that the proposed CAAF outperforms the state-of-the-art forwarding strategy regarding content retrieval delay and Interest satisfaction ratio across diverse scenarios.

Index Terms—Vehicular ad-hoc networks, named data networking, forwarding strategy, cache prediction.

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I. INTRODUCTION

DURING the last couple of decades, Vehicular Ad-hoc Networks (VANETs) have gained increasing attention from both academia and industry [1], [2], [3]. As vehicles may frequently change locations, VANETs are characterized by short-duration and interruption-prone links with rapidly fluctuating network conditions [4]. Nevertheless, traditional host-centric addressing and connection-oriented content delivery necessitate the establishment of end-to-end connections before data transmission [5]. Their connection-based communication paradigm is ill-suited for content delivery in VANETs, resulting in low transmission efficiency. VANETs urgently call for a novel network architecture that aligns with their inherently dynamic nature.

Fortunately, Named Data Networking (NDN) [6], a prominent future network architecture, shifts the communication paradigm from host-centric to content-centric [7], making it particularly suitable for VANETs. Specifically, NDN utilizes content name instead of IP address to uniquely identify content, decoupling each content from its source location [8]. Additionally, nodes in NDN are inherent with cache ability, which enables consumers to retrieve content not only from providers but also from cached replicas at any intermediate node. This feature enables NDN to mitigate the dynamic topology and transitory connectivity issues in VANETs [9]. Moreover, researchers have provided strategies from multiple aspects such as transmission optimization [10], [11], [12] and security safeguard [13], [14], [15] to ensure the transmission efficiency of NDN networks. Given these benefits, Grassi et al. [16] first cleverly integrated Named Data Networking (NDN) and VANETs to create NDN-VANETs.

Although NDN-VANETs present numerous advantages, an effective forwarding strategy that guarantees reliable transmission under highly dynamic VANETs is still missing. Specifically, the basic forwarding strategy in NDN-VANETs is flooding, where each vehicle rebroadcasts the Interest packet it has received, generating a large amount of redundancy. While some strategies reduce redundancy through passive suppression-based forwarding [17], [18], they can not guarantee reliability since they lack awareness of the state of the nodes in the network. Moreover, many existing forwarding strategies [19], [20] primarily focus on how to select “better” intermediate nodes to reach the content providers, neglecting the latest changes of in-network caching. Therefore, they fail to efficiently utilize the

in-network caching capabilities provided by NDN networks. In most cases, consumers still need to transmit their requests to the original content providers to obtain the required content, which not only increases the transmission latency but also reduces the satisfaction ratio. Besides, this long path of transmission from the consumer to the original provider also makes it difficult to ensure reliability.

In this paper, we propose a novel NDN-based cache-aware adaptive forwarding (CAAF) strategy for VANETs. CAAF achieves reliable content retrieval by proactively exploring in-network caching and adaptively selecting forwarding nodes with consideration of fast delivery and stable transmission. It consists of three modules: cache prediction module, responsible for real-time prediction of the in-network caching; filtering module, which filters out intermediate nodes that cannot meet the transmission requirements; and decision module, which decides the best forwarding node within the candidate nodes. Specifically, CAAF operates as follows: First, as a core component of the cache prediction module, a content information table is designed for each vehicle to record information about the Interest packets it receives. These tables are updated periodically across all vehicles and a prediction model is used to predict real-time in-network caching during the update interval. Second, a coarse-grained filtering mechanism is developed in the filtering module to sieve candidate forwarding vehicles that satisfy both the stability and accessibility requirements. Finally, within these candidate vehicles, CAAF utilizes a multi-attribute decision-making approach, which comprehensively considers factors including distance, transmission delay, and satisfaction rate, to determine the optimal forwarding node. To evaluate the performance of CAAF strategy, we conduct extensive simulations across diverse scenarios. The results validate the accuracy of the proposed prediction model and demonstrate the superiority of CAAF compared to state-of-the-art forwarding strategies in NDN-VANETs.

The contributions of the paper are summarized as follows:

- We design a proactive cache exploration method, which consists of periodic cache information retrieval and continuous cache prediction. The cache prediction model utilizes historical information and linear prediction to achieve global cache awareness in real time.
- We develop a filtering mechanism to sieve candidate vehicles aligning with the requirements of reliable transmission, then use a multi-attribute decision-making method to determine the optimal one. This approach correctly evaluates the performance of individual nodes and enhances adaptability in different environments.
- We conducted extensive simulations compared with existing strategies to evaluate the performance of the proposed CAAF strategy. The results show that CAAF outperforms existing strategies in terms of transmission redundancy, content retrieval delay, and Interest satisfaction ratio.

The remainder of this paper is structured as follows. First, Section II summarizes the related work. Then, Section III describes the system model. The complete forwarding strategy consisting of three modules is presented in Section IV. Finally,

the performance evaluation and analysis are conducted in Section V, and conclusions are drawn in Section VI.

II. RELATED WORK

To prevent broadcast redundancy and reverse path partitioning, extensive research has been conducted on forwarding strategies in NDN-VANETs. According to the forwarding strategy, existing methods can be categorized into two types: passive suppression-based forwarding and proactive selection-based forwarding.

1) *Passive suppression-based Forwarding*: The passive suppression-based forwarding strategies set different priorities for each node and passively suppress the forwarding of low-priority nodes through a delay timer. Typically, the priority assigned to a node is based on factors such as distance and link stability. Grassi et al. [21] and Yu et al. [22] employed a delay timer mechanism to prioritize vehicles further from the current sender in packet forwarding, respectively. To further enhance efficiency, Kuai et al. [23] and Rondon et al. [24] considered node density and road characteristics when assigning node priority, respectively. Li et al. [25] improved the content retrieval success ratio by introducing a suppression angle, where among vehicles receiving duplicate Interests, only those within the suppression angular scope are inhibited from forwarding, preserving opportunities for others on viable paths. Furthermore, to overcome reverse path partitioning caused by vehicles' movement, Boukerche et al. [26] set the delay timer inversely proportional to the link expiration time to improve the link stability. Similarly, Zafar et al. [27] proposed a distributed receiver-driven protocol that prioritizes nodes with higher link stability for forwarding. While these passive suppression-based forwarding strategies are simple to implement, they tend to create a lot of redundancy and introduce additional latency.

2) *Proactive selection-based Forwarding*: The proactive selection-based forwarding strategies rely on proactively selecting forwarding nodes from neighbors based on multiple evaluation metrics. Wang et al. [28] applied Long Short-Term Memory (LSTM) to predict the Neighbor Table (NBT) for precise neighboring vehicles' positions, subsequently selecting forwarding nodes based on the predicted NBT. Li et al. [29] designed a dual-pronged strategy: an optimal single-path forwarding strategy for Interest packets and a reliable multi-path backup forwarding strategy for data packets. Complementarily, Guo et al. [30] proposed a Bayesian-based Receiver Forwarding Decision (BRFD) strategy, adapting to real-time network conditions obtained through neighbor interactions. Rui et al. [31] proposed a strategy to build multiple stable unicast forwarding paths in NDN-VANETs. Furthermore, Hou et al. [32] segmented urban roads into complex multi-junction and straight-lane scenarios, then choreographed Data packet forwarding in harmony with vehicles' mobility patterns to improve adaptability to consumer movements. Although proactive selection-based forwarding strategies suppress redundancy, it is still debatable how to exactly select the optimal nodes.

The existing research primarily focuses on controlling the number of forwarding nodes to suppress redundancy and

TABLE I
COMPARISON OF RELATED WORK AND OUR WORK

Strategies	Core Idea	Advantages	Shortcomings
passive suppression based [21], [22], [27]	set different priorities for each node and passively suppress the forwarding of low-priority nodes	- reduce a certain amount of redundancy - easy to implement	- cannot completely resolve redundancy - introduces additional latency
proactive selection based [28], [29], [32]	proactively selecting forwarding nodes from neighbors based on multiple evaluation metrics	- almost complete resolution of redundancy - more precise selection of forwarding nodes	- no utilization of in-network caching - inadequate adaptation to network state changes
our proposed strategy	predict in-network caching and adaptively select forwarding nodes that prioritize fast delivery and stable transmission	- efficiently utilize in-network caching - adaptation to network state changes	- need for roadside units - additional communication overhead (negligible)

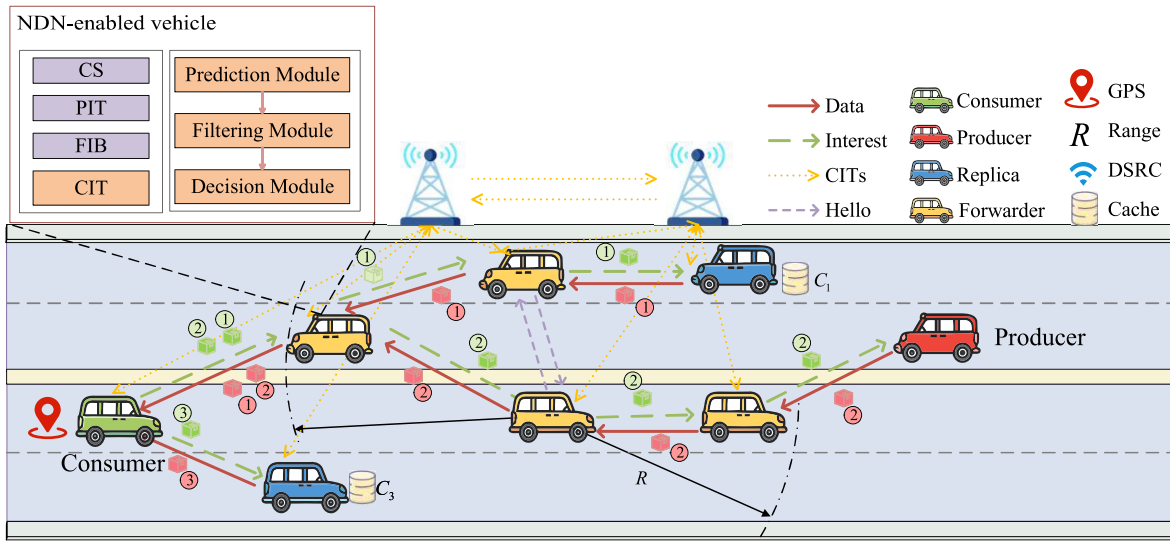


Fig. 1. System architecture of CAAF.

selecting more suitable forwarding nodes to enhance transmission efficiency. However, there is still a lack of integrated forwarding strategies that leverage in-network caching, and the parameters considered during the forwarding decision process are insufficient. Therefore, this paper proposes a cache-aware adaptive forwarding strategy. By predicting in-network caching and employing multi-attribute decision-making, our approach can forward requests to the optimal node, thereby reducing content retrieval delay and improving content retrieval success rate. The summary of related work and our work is shown as Table I.

III. PRELIMINARY

In this section, we first illustrate the network model of cache-aware adaptive forwarding (CAAF) strategy and then introduce the details of the cache replacement policy and the design of the content information table.

A. Network Model

The network model of CAAF is depicted in Fig. 1, including both mobile vehicles and stationary Road Side Units (RSUs).

Denote vehicles within network as $\mathcal{V} = \{V_1, V_2, \dots, V_N\}$, where $N = |\mathcal{V}|$ represents the total number of vehicles. Each vehicle is equipped with a Global Positioning System (GPS) to obtain its real-time mobility information, including location, velocity, and direction. This mobility information is periodically exchanged with neighboring vehicles to assist in making forwarding decisions. Consistent with the characteristics of NDN, each vehicle is equipped with a cache. Without loss of generality, we assume that all vehicles have the same cache capacity of s content items. Communication between vehicles primarily relies on wireless interfaces compliant with Dedicated Short Range Communication (DSRC) standards, such as IEEE 802.11p. Considering the finite range restrictions of wireless communication, we assume all vehicles share an equal maximum communication distance, represented as R .

Each vehicle can play one of four roles:

- *Consumer*: A vehicle that sends Interest packets to request specific content from the network.
- *Content Provider*: A vehicle that owns the original content data and can generate corresponding Data packets to satisfy the Interest packet requests.

- *Replica Content Source*: A vehicle that is not the original Provider but has cached replica of some content, so can also generate corresponding Data packets upon receiving Interest packet requests.
- *Forwarder*: A vehicle that is just responsible for forwarding Interest and Data packets based on the forwarding strategy.

Since both the Content Provider and Replica Content Source contain requested content, we refer to them collectively as content sources, which means that the same request can be satisfied by multiple nodes. Each vehicle contains the three data structures of NDN: Content Store (CS), Pending Interest Table (PIT), and Forwarding Information Table (FIB). Besides, we design a new content information table for each vehicle to record information about the Interest packets it receives. These information can be used to infer the cached content on each vehicle. We use RSUs to collect these tables and disseminate them among all the vehicles. Meanwhile, to reduce the substantial burden of constant real-time communications, the collection and dissemination operations are executed periodically at intervals of T time.

B. Cache Replacement Policy

Given the finite cache capacity, when the cache storage becomes full, it will execute a cache replacement policy to determine which content should be evicted to make room for the new ones [33]. As a simple but classic cache replacement policy, Least Recently Used (LRU) has been recognized as the most common replacement policy in web caching and is also the default cache replacement policy in NDN [34], [35]. Consequently, we assumed that vehicles within the network employ the LRU cache replacement policy. The primary principle underlying the LRU cache replacement policy is to remove the content that has not been requested for the longest time when the cache reaches its capacity, making room for caching the latest content. Conversely, when the cache has available capacity, it readily caches any requested content without any deletions.

Generally, the consumers request a catalog of content items, denoted as $\mathcal{C} = \{C_1, C_2, \dots, C_M\}$, where $M = |\mathcal{C}|$ is the total number of content items. Let $\mathcal{T}(t) = \{\tau_1(t), \tau_2(t), \dots, \tau_M(t)\}$, where $\tau_m(t)$ represents the last request time of C_m till time t . Arranging the $\mathcal{T}(t)$ in descending order, we can obtain the order of C_m , denoted as $o_m(t) \in \{1, 2, \dots, M\}$. We use $y_m(t)$ to indicate whether C_m is cached at time t , which takes binary values: 1 means the item is stored and 0 means it is not. Since the capacity of the cache is s , then the cache state under LRU can be illustrated as:

$$y_m(t) = \begin{cases} 1, & o_m(t) \leq s, \\ 0, & o_m(t) > s. \end{cases} \quad (1)$$

Consequently, we can deduce that the contents cached at time t are the top s contents with the largest last requested time till time t .

It is worth noting that we use the LRU policy here because it is simple and performs well enough. Moreover, if other cache replacement policies need to be used in some scenarios, the

TABLE II
CONTENT INFORMATION TABLE

Content	Request times	Last request time	Arrival rate
C_1	k_1^1	τ_1^1	λ_1^1
C_2	k_1^2	τ_1^2	λ_1^2
$\dots$	$\dots$	$\dots$	$\dots$

proposed strategy in this paper can be easily adapted with readily change of the prediction model.

C. Content Information Table

To track the cached content of each vehicle, we add a new data structure called Content Information Table (CIT) in addition to the traditional NDN structures for each vehicle. $CIT_i(t)$ records the requested contents' information on V_i until time t . The structure of the CIT of V_i is shown in Table II, where each entry contains four parameters:

- *requested content*: the specific requested content, denoted as C_m ;
- *requested times*: the total number of requests for C_m received by V_i , denoted as k_i^m ;
- *last request time*: the latest arrival time of the request for C_m at V_i , denoted as τ_i^m ;
- *arrival rate*: the arrival rate of requests for C_m at V_i , denoted as λ_i^m .

Assuming that the arrival process of requests at V_i obeys a Poisson process with arrival rate λ_i . It can be calculated as $\lambda_i = N_i/t$, where N_i represents the number of requests received by V_i till t . Then each request arrival process for content C_m obeys a Poisson process with arrival rate λ_i^m [36], which satisfies:

$$\lambda_i^m = \lambda_i \cdot \frac{k_i^m}{\sum_{C_m \in \mathcal{C}} k_i^m}, \quad (2)$$

where the parameters k_i^m and τ_i^m can be obtained from each request directly, while λ_i^m can be obtained according to (2). Every T seconds, each vehicle uploads its CIT to the RSU. After receiving the tables from all vehicles, the RSU aggregates them and distributes the aggregated table back to all vehicles. Upon receiving content requests from other vehicles, a vehicle is tasked with updating the corresponding entries in its CIT to reflect the latest request information. Those important notations used throughout this paper are summarized in Table III.

D. Design Principles

While there have been numerous studies on forwarding strategies in NDN-VANETs, most of them overlook the effective utilization of in-network caching. This limitation prevents these strategies from significantly reducing content retrieval delay and improving the content retrieval success rate. In highly dynamic VANET environments, there is still a lack of solutions that can accurately locate and efficiently utilize cached content. Therefore, the goal of this paper is to enable nodes with cache awareness and enhance transmission efficiency by leveraging in-network caching. To achieve this goal, two key tasks need to be addressed:

TABLE III
NOTATIONS USED THROUGHOUT THE PAPER

Notation	Description
R	The max communication distance between vehicles
V_i	The i -th vehicle
C_m	The m -th content item
t	The current time
s	The cache capacity of each vehicle
T	The distributed period of CITs
k_i^m	The total number of requests for C_m received by V_i
τ_i^m	The latest arrival time of the request for content C_m at V_i
λ_i^m	The arrival rate of requests for C_m at V_i
$\hat{P}_i^m(t)$	The predicted probability that V_i had cached C_m at time t
P_{th}	The threshold that determines whether a content is cached.
$L_{ij}(t)$	The LET between V_i and V_j at time t
$G(i)$	The intersection region formed by V_i and content sources
d_j	The distance attribute of V_j
l_j	The transmission latency attribute of V_j
r_j	The transmission success ratio attribute of V_j
$\mathcal{N}(i)$	The candidate set of next forwarding vehicle nodes of V_i
$F(i)$	The decided next forwarding vehicle node of V_i

1) *Real-time in-network cache awareness*: A new data structure, CIT, has been added for each vehicle node, enabling awareness of in-network caching through periodic interactions. However, this periodic awareness is insufficient to reflect the real-time cache state due to the possibility of cache replacement and new requests. Consequently, prediction algorithms must be designed to obtain a more accurate real-time cache state.

2) *Optimal forwarding node selection*: Based on accurate cache awareness, the next task is to select an intermediate node as a forwarding relay to efficiently reach the content source. To achieve this, we must design a decision-making algorithm that evaluates multiple metrics of candidate nodes (e.g., distance, delay, and transmission success rate) to identify the optimal node.

Next, we describe our proposed forwarding strategy in detail, which addresses both of these tasks well.

IV. THE PROPOSED STRATEGY

Our proposed CAAF strategy consists of three modules as illustrated in Fig. 1. First, a prediction module is responsible for predicting the cache contents of each vehicle in real-time, thus enabling the cache awareness of the vehicle. Second, a filtering module filters out the intermediate nodes that cannot satisfy the transmission requirements, forming a set of candidate forwarding nodes. Third, a decision module decides the optimal forwarding node from the candidate set; Together, these three modules allows the most accurate forwarding decision according to the network state improving the transmission efficiency.

A. Prediction Module

RSUs collect and disseminate these tables among all the vehicles at intervals of T time. To reduce the overhead of real-time updates and improve the accuracy of the information, we propose a prediction model to predict the cached content on each vehicle during the update interval.

Denote the probability that C_m has been cached on V_i at time t as $P_i^m(t)$. Given that the interval for cache information

updating is T , we can leverage the distributed cache information $CIT(nT)$ from the n th period to predict $P_i^m(t) \in [0, 1]$ at time $t = nT + \tau, 0 \leq \tau \leq T$.

Let $\mathcal{C}_i(nT)$ denote the content set cached on the vehicle V_i at time nT , we can partition this set into two categories based on their latest requested time. The first category consists of contents whose latest request times are greater than that of content C_m , denoted as $\mathcal{C}_i^+(nT)$. The second category contains those contents whose latest requested times are less than that of content C_m or have not been requested yet, denoted as $\mathcal{C}_i^-(nT)$. To simplify, we omit nT from the notation. If the content C_m was already cached on vehicle V_i at time nT , i.e., $O_i^m(nT) \leq S$, then the content C_m would remain cached on vehicle V_i at time t under the following two possible conditions:

- 1) During the interval $nT < t < nT + \tau$, the number of requests for different contents in $\mathcal{C}_i^-$ are less than $S - |\mathcal{C}_i^+|$. Under this condition, the cache on vehicle V_i would not have filled up enough to evict content C_m based on the LRU policy.
- 2) During the interval $nT < t < nT + \tau$, although the number of requests for different contents in $\mathcal{C}_i^{m-}$ surpasses $S - |\mathcal{C}_i^+|$, content C_m still holds a high enough order, i.e., $O_i^m(t) \leq S$. This implies that even though the cache has reached its capacity, content C_m was still among the most recently requested, hence retaining its place in the cache based on the LRU policy.

The probability under the first condition, denoted as $P_{i,m}^1(t)$, can be calculated as:

$$P_{i,m}^1(t) = P \left(\sum_{C_u \in \mathcal{C}_i^-} \mathbb{I}(N_i^u(\tau)) \leq S - |\mathcal{C}_i^+| \right), \quad (3)$$

where $N_i^u(\tau)$ represents the number of requests for C_u arrived at V_i within $[nT, nT + \tau]$ and $\mathbb{I}(N_i^u(\tau))$ is an indicator function which satisfies:

$$\mathbb{I}(N_i^u(\tau)) = \begin{cases} 1, & N_i^u(\tau) \geq 1 \\ 0, & N_i^u(\tau) = 0 \end{cases}. \quad (4)$$

Defining the variable $X_i^u(\tau) = 1 - \mathbb{I}(N_i^u(\tau))$ and substitute it to (3), we obtain:

$$\begin{aligned} P_{i,m}^1(t) &= P \left(\sum_{u \in \mathcal{C}_i^-} (1 - X_i^u(\tau)) \leq S - |\mathcal{C}_i^+| \right) \\ &= P \left(|\mathcal{C}_i^-| - \sum_{u \in \mathcal{C}_i^-} X_i^u(\tau) \leq S - |\mathcal{C}_i^+| \right) \\ &= P \left(\sum_{u \in \mathcal{C}_i^-} X_i^u(\tau) \geq |\mathcal{C}_i^-| - S \right). \end{aligned} \quad (5)$$

Since $X_i^u(\tau)$ takes the value 1 if there are no requests ($N_i^u(\tau) = 0$) and 0 otherwise, it obvious that it follows a Bernoulli distribution. Since the arrival of requests for C_u on vehicle V_i obeys a Poisson process with parameter λ_i^u , then we

can deduce that:

$$P(N_i^u(\tau) = 0) = e^{-\lambda_i^u \tau}, \quad (6)$$

$$P(N_i^u(\tau) \geq 1) = 1 - e^{-\lambda_i^u \tau}. \quad (7)$$

Therefore $X_i^u(\tau)$ follows a Bernoulli distribution with mean $\mu = e^{-\lambda_i^u \tau}$, i.e., $X_i^u(\tau) \sim \text{Bernoulli}(e^{-\lambda_i^u \tau})$.

Theorem 1: Define $S(\tau) = \sum_{u \in \mathcal{C}_i^-} X_i^u(\tau)$, then S obeys a Gaussian distribution, i.e., $S_i(\tau) \sim \mathcal{N}(\mu_i, \sigma_i^2)$, where $\mu_i = \sum_{u \in \mathcal{C}_i^-} e^{-\lambda_i^u \tau}$ and $\sigma_i^2 = \sum_{u \in \mathcal{C}_i^-} e^{-\lambda_i^u \tau} (1 - e^{-\lambda_i^u \tau})$.

Proof: Please see details in Appendix, available online. $\square$

According to Theorem 1, the probability in (5) can be derived as:

$$\begin{aligned} P_{i,m}^1(t) &= P(S_i \geq |\mathcal{C}_i| - S) \\ &= P\left(\frac{S_i - \mu_i}{\sigma_i} \geq \frac{|\mathcal{C}_i| - S - \mu_i}{\sigma_i}\right) \\ &= \frac{1}{2} \text{erfc}\left(\frac{(|\mathcal{C}_i| - S - \mu_i)/\sigma_i}{\sqrt{2}}\right), \end{aligned} \quad (8)$$

where $\text{erfc}(x)$ denotes the complementary Gaussian error function, represented as $\text{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^{+\infty} e^{-t^2} dt$.

Next, the probability under the second condition denoted as $P_{i,m}^2(t)$, can be formulated as:

$$\begin{aligned} P_{i,m}^2(t) &= P\left(\sum_{u \in \mathcal{C}_i^-} \mathbb{I}(N_i^u(\tau)) > S - |\mathcal{C}_i^+|, \right. \\ &\quad \left. \sum_{u \in \mathcal{C}_i^-} \mathbb{I}(T_u) < S - |\mathcal{C}_i^+| \right), \end{aligned} \quad (9)$$

where T_u represents the latest requested time of content C_u , and $\mathbb{I}(T_u)$ satisfies:

$$\mathbb{I}(T_u) = \begin{cases} 1, & T_u \geq T_m, \\ 0, & T_u < T_m. \end{cases} \quad (10)$$

Since solving $P_{i,m}^2$ requires considering the number of requests and arrival times of all contents in $\mathcal{C}_i^-$, it is hard to obtain an exact value of $P_{i,m}^2$. Therefore, we only consider the worst case under this scenario to derive its lower bound, denoted as $P_{i,m}^{2-}$.

The worst case assumes that the number of requests for different content C_u in $\mathcal{C}_i^-$ and content C_m are both only one, and the requested time for content C_m is the latest. Then we can formulate $P_{i,m}^{2-}$ by:

$$P_{i,m}^{2-} = P(T_m > T_u, N_i^m(\tau) = N_i^u(\tau) = 1, \forall u \in \mathcal{C}_i^-). \quad (11)$$

Theorem 2: Given that the number of requests for content C_m within the time intervals t_1 and t_2 is 1, then its arrival time follows a uniform distribution within t_1 and t_2 , i.e., $T_m \sim \mathcal{U}(t_1, t_2)$.

Proof: Please see the Appendix for details, available online. $\square$

Proposition 1: The probability that the requested time for content C_m is larger than content C_u in $\mathcal{C}_i^-$ equals to $\frac{1}{2}$, i.e.,

$$P(T_m > T_u) = \frac{1}{2}, \forall u \in \mathcal{C}_i^-. \quad (12)$$

Proof: Please see the Appendix for details, available online. $\square$

According to Theorem 2 and Proposition 1, we can further solve the probability in (11) as:

$$\begin{aligned} P_{i,m}^{2-}(t) &= P(N_i^m(\tau) = 1) \prod_{u \in \mathcal{C}_i^-} P(N_i^u(\tau) = 1) P(T_m > T_u) \\ &= \lambda_i^m e^{-\lambda_i^m \tau} \prod_{u \in \mathcal{C}_i^-} \frac{1}{2} \lambda_i^u e^{-\lambda_i^u \tau} \\ &= \left(\frac{1}{2}\right)^{|\mathcal{C}_i^-|} e^{-(\lambda_i^m + \sum_{u \in \mathcal{C}_i^-} \lambda_i^u) \tau} \lambda_i^m \prod_{u \in \mathcal{C}_i^-} \lambda_i^u, \end{aligned} \quad (13)$$

where $\lambda_i^- = \sum_{u \in \mathcal{C}_i^-} \lambda_i^u \tau$.

Since $P_i^m = P_{i,m}^1 + P_{i,m}^2$, we can conclude that:

$$\begin{aligned} \inf P_i^m(t) &= P_{i,m}^1(t) + P_{i,m}^{2-}(t) \\ &= \frac{1}{2} \text{erfc}\left(\frac{|\mathcal{C}_i| - S - \mu_i}{\sqrt{2} \sigma_i}\right) \\ &\quad + \left(\frac{1}{2}\right)^{|\mathcal{C}_i^-|} e^{-(\lambda_i^m + \sum_{u \in \mathcal{C}_i^-} \lambda_i^u) \tau} \lambda_i^m \prod_{u \in \mathcal{C}_i^-} \lambda_i^u. \end{aligned} \quad (14)$$

Similarly, if the content C_m was not cached on vehicle V_i at time nT , we only consider the worst scenario that allow C_m to be cached on V_i at time t , which is composed of the following two conditions:

- 1) The number of requests for C_m during the time interval $(nT, nT + \tau)$ is greater than 0, while the number of requests for each different C_u , $u \neq m$ is 0.
- 2) The number of requests for each content in $\mathcal{C}$ during the time interval $(nT, nT + \tau)$ is 1, and the request time of C_m is the latest.

Denote the probabilities of these two conditions as $P_{i,m}^3(t)$ and $P_{i,m}^4(t)$, we can calculate them as follows:

$$\begin{aligned} P_{i,m}^3(t) &= P(N_i^m(\tau) \geq 1) \prod_{u \neq m} P(N_i^u(\tau) = 0) \\ &= (1 - e^{-\lambda_i^m \tau}) \prod_{u \neq m} e^{-\lambda_i^u \tau} \\ &= (1 - e^{-\lambda_i^m \tau}) e^{-\sum_{u \neq m} \lambda_i^u \tau} \\ &= e^{-\lambda_i \tau} (e^{\lambda_i^m \tau} - 1). \end{aligned} \quad (15)$$

$$\begin{aligned} P_{i,m}^4(t) &= P(N_i^m(\tau) = 1) \prod_{u \neq m} P(N_i^u(\tau) = 1) P(T_m > T_u) \\ &= \lambda_i^m e^{-\lambda_i^m \tau} \prod_{u \neq m} \frac{1}{2} \lambda_i^u e^{-\lambda_i^u \tau} \end{aligned}$$

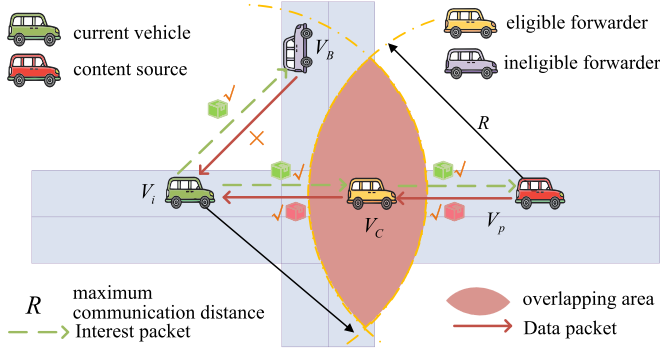


Fig. 2. Eligible forwarders.

$$= \left(\frac{1}{2}\right)^M e^{-\lambda_i \tau} \prod_{k \in C} \lambda_i^k. \quad (16)$$

Thus, we can derive the lower bound of $P_i^m(t)$ as:

$$\begin{aligned} \inf P_i^m(t) &= P_{i,m}^3(t) + P_{i,m}^4(t) \\ &= e^{-\lambda_i \tau} (e^{\lambda_i^m \tau} - 1) + \left(\frac{1}{2}\right)^M e^{-\lambda_i \tau} \prod_{k \in C} \lambda_i^k. \end{aligned} \quad (17)$$

For a conservative estimate of the caching probability, we set the probability that content C_m is cached on the vehicle V_i at time t to its lower bound, i.e., $\hat{P}_i^m(t) = \inf P_i^m(t)$. Then the predicted probability can be represented by

$$\hat{P}_i^m(t) = \begin{cases} e^{-\lambda_i \tau} (e^{\lambda_i^m \tau} - 1) \\ \quad + \left(\frac{1}{2}\right)^M e^{-\lambda_i \tau} \prod_{k \in C} \lambda_i^k, & O_i^m(nT) > s, \\ \left(\frac{1}{2}\right)^{|C_i^-|} e^{-(\lambda_i^m + \lambda_i^-) \tau} \prod_{u \in C_i^-} \lambda_i^u \\ \quad + \frac{1}{2} \operatorname{erfc} \left(\frac{|C_i^-| - S - \mu_i}{\sqrt{2} \sigma_i} \right), & O_i^m(nT) \leq s. \end{cases} \quad (18)$$

When a vehicle receives an Interest packet, it accesses the content name from the packet and extracts the relevant information from its CIT. Then, it uses this information to calculate the caching probability based on (18), which serves as the output of the prediction module. This prediction paves the way for identifying potential content sources — vehicles with a $\hat{P}_i^m(t)$ value surpassing a predetermined threshold, P_{th} .

B. Filtering Module

Initially, upon obtaining the predicted content sources, we can facilitate multi-hop forwarding to these sources to retrieve the desired content. However, due to the varying positions and mobility of vehicles, not all vehicles are feasible as relay nodes to reach these sources. Consequently, there is a necessity to identify nodes that are suitably positioned and stable enough to serve as relay nodes for content delivery.

First, the current geographical distribution of vehicles must be taken into account to meet the accessibility requirement. As depicted in Fig. 2, V_i is the current vehicle sending an Interest packet to require content, while V_p is the content source vehicle capable of satisfying the corresponding Interest request. V_B and

V_C are both neighbor nodes within the communication range of V_i , thus capable of receiving Interest packets forwarded by V_i and being the next hop. However, only V_C is located between V_i and V_p , while V_B is outside this area. If V_i chooses V_B as the next forwarding vehicle, it won't be able to forward the Interest packet to V_p . The Interest packet only has a chance of being successfully forwarded to the content source V_p when V_C is selected as the next forwarding vehicle. Consequently, only those vehicles located within the pink overlapping area, denoted as Z_i^p , which represents the region between the current vehicle V_i and the targeted content source V_p , are eligible candidates for acting as intermediate nodes to reach the content source.

Second, to avoid the issue of reverse path partitioning, it is imperative to consider Link Expiry Time (LET), which refers to the duration a communication link remains stable before it potentially breaks due to node mobility. It is an important indicator of link stability as links with longer LETs can exist longer, thus ensuring better sustainability throughout the content retrieval process.

Proposition 2: Assuming that V_i is located at position $(x_i(t), y_i(t))$ with velocity of $(v_i^x(t), v_i^y(t))$, and V_j is located at position $(x_j(t), y_j(t))$ with velocity of $(v_j^x(t), v_j^y(t))$ at time t . Define the LET between V_i and V_j at time t as $L_{ij}(t)$, then

$$L_{ij}(t) = \frac{-(ab + cd) + \sqrt{(a^2 + c^2)R^2 - (ad - bc)^2}}{a^2 + c^2}, \quad (19)$$

where $a = v_i^x(t) - v_j^x(t)$, $b = x_i(t) - x_j(t)$, $c = v_i^y(t) - v_j^y(t)$, $d = y_i(t) - y_j(t)$.

Proof: Please see the Appendix for details, available online. $\square$

Given that the time required to complete a content retrieval process is represented by the Round Trip Time (RTT), a necessary condition for successful packet transmission is that the LET must exceed the RTT. This condition guarantees that the link does not expire before the Data packet returns, thereby avoiding potential interruptions or failures in the communication process.

Based on the above considerations, we can identify vehicles that are well-positioned and stable enough to serve as intermediate nodes for reaching content sources. These vehicles constitute a candidate set, which can be denoted as:

$$\mathcal{N}(i) = \{V_j | L_{ij}(t) > RTT_i(t), j \in G(i)\}, \quad (20)$$

where $G(i) = Z_i^{p_1} \cup Z_i^{p_2} \dots \cup Z_i^{p_n}$ denotes the union of all the intersection regions $Z_i^{p_k}$ formed by vehicle V_i and content source V_{p_k} . The computation of (20) requires location information for these content sources, which is available through periodic information exchange. In our implementation, this location information is appended to their CITs, which are regularly collected and synchronized through RSUs.

C. Decision Module

Post the assembly of candidate vehicles, considering each one as the forwarder would result in unnecessary redundancy. Hence, it's imperative to select the optimal forwarding vehicle among them. To accomplish this, we first introduce key metrics

to evaluate each vehicle's performance and then devise a multi-attribute decision-making method to determine the vehicle with the best overall performance across these metrics.

1) *Evaluation Attributes*: To select the optimal relay node from $\mathcal{N}(i)$ for forwarding, we base our decision on three critical attributes, described as follows:

- *Distance*: This attribute refers to the distance along the road between the current vehicle V_i and neighbor vehicle V_j , aiming to reduce hop count in transmissions. It is denoted as d_j and computed as:

$$d_j = E_{ij} \cdot \cos \left(\alpha - \arctan \left| \frac{x_j - x_i}{y_j - y_i} \right| \right), \quad (21)$$

where E_{ij} is the euclidean distance between V_i and V_j , i.e., $E_{ij} = \sqrt{(x_i(t) - x_j)^2 + (y_i - y_j)^2}$, and α is the road direction.

- *Transmission Latency*: This attribute refers to the average delay of content delivery through V_j , indicating the efficiency of content delivery. It is denoted as l_j and calculated as:

$$l_j = t_r - t_s, \quad (22)$$

where t_s represents the time of forwarding the Interest packet, while t_r represents the time of receiving the corresponding Data packet.

- *Transmission Success Ratio*: This attribute refers to the ratio of satisfied Interest requests, which is a vital metric assessing content delivery effectiveness. It is denoted as r_j and calculated as:

$$r_j = \frac{R_j}{N_j}, \quad (23)$$

where R_j and N_j represent the satisfied and total number of requested Interests on V_j respectively.

2) *Optimal Forwarding Vehicle Selection*: Then we need to select the optimal node from $\mathcal{N}(i)$ according to these attributes. To do this, we first construct it as a multi-objective optimization problem:

$$\max_{V_j \in \mathcal{N}(i)} \mathbf{U}(j) = \max_{V_j \in \mathcal{N}(i)} [U_d(j), U_l(j), U_r(j)], \quad (24)$$

where $\mathbf{U}(j) = [U_d(j), U_l(j), U_r(j)]$ is the vector of attributes, $U_d(j)$, $U_l(j)$ and $U_r(j)$ denote the objective utility function of V_j in the three attributes respectively. The objective is to optimize all the attributes in $\mathbf{U}(j)$. To solve this optimization problem, we apply a multi-attribute decision-making method, Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) [37]. The steps involved in the method are described as follows, to simplify, we omit the subscript j for each attribute in the following.

Step 1: Homogenization of Attributes: Recognizing that different attributes have varying directional preferences— d and r preferring larger values to denote better performance, whereas l favors lower values, we first execute a homogenization process to ensure that all attributes meet the same requirements for large

values. Specifically, we incrementally adjust the l attribute by

$$l' = \frac{1}{l}. \quad (25)$$

This essential step harmonizes the disparate orientation of these attributes to uniformly reflect the best possible scenario across all attributes. Then we can obtain the initial decision matrix illustrated as:

$$\mathbf{X} = \begin{bmatrix} & \mathbf{d} & \mathbf{l}' & \mathbf{r} \\ V_1 & x_{(1,1)} & x_{(1,2)} & x_{(1,3)} \\ V_2 & x_{(2,1)} & x_{(2,2)} & x_{(2,3)} \\ \vdots & \vdots & \vdots & \vdots \\ V_n & x_{(n,1)} & x_{(n,2)} & x_{(n,3)} \end{bmatrix}, \quad (26)$$

where $\{V_1, V_2, \dots, V_n\}$ represents the candidate vehicles in $\mathcal{N}(i)$, $\{\mathbf{d}, \mathbf{l}', \mathbf{r}\}$ represents the corresponding attributes, and $x_{(j,k)}$ represents the k th attribute of the j th vehicle.

Step 2: Construct normalized matrix: Given the potential for scale discrepancies among attributes, which can impede decision accuracy, it's crucial to normalize the matrix. The elements of matrix $\mathbf{X}$ are normalized by:

$$z_{(j,k)} = \frac{x_{(j,k)}}{\sqrt{\sum_{j=1}^n x_{(j,k)}^2}}. \quad (27)$$

Then we can obtain the normalized decision matrix as $\mathbf{Z}$. The rows of the matrix represent each alternative solution, denoted as $\vec{h}_j = [z_{(j,1)}, z_{(j,2)}, z_{(j,3)}]$, while the columns represent each attribute, denoted as $\vec{p}_k = [z_{(1,k)}, z_{(2,k)}, \dots, z_{(n,k)}]$.

Step 3: Positive Ideal and Negative Ideal Solutions: The next step is to determine the positive ideal solution $\vec{h}^+$ and negative ideal solution $\vec{h}^-$. The positive ideal solution represents the best possible performance of each attribute, while the negative ideal solution represents the worst performance. They are calculated as:

$$\vec{h}^+ = (p_1^+, p_2^+, p_3^+), \quad \vec{h}^- = (p_1^-, p_2^-, p_3^-), \quad (28)$$

where $p_k^+ = \max\{z_{(1,k)}, z_{(2,k)}, \dots, z_{(n,k)}\}$ and $p_k^- = \min\{z_{(1,k)}, z_{(2,k)}, \dots, z_{(n,k)}\}$ represent the highest and lowest value of attribute $\vec{p}_k$ among all candidate vehicles, respectively.

Step 4: Similarity Calculation: Then, we calculate the Euclidean distance between each alternative solution and the two ideal solutions by:

$$\begin{aligned} D_j^+ &= \|\vec{h}_j - \vec{h}^+\| = \sqrt{\sum_{k=1}^3 (p_k^+ - z_{(j,k)})^2}, \\ D_j^- &= \|\vec{h}_j - \vec{h}^-\| = \sqrt{\sum_{k=1}^3 (p_k^- - z_{(j,k)})^2}. \end{aligned} \quad (29)$$

Consequently, the similarity score of each vehicle, denoted as S_j , is calculated by:

$$S_j = \frac{D_j^-}{D_j^+ + D_j^-}. \quad (30)$$

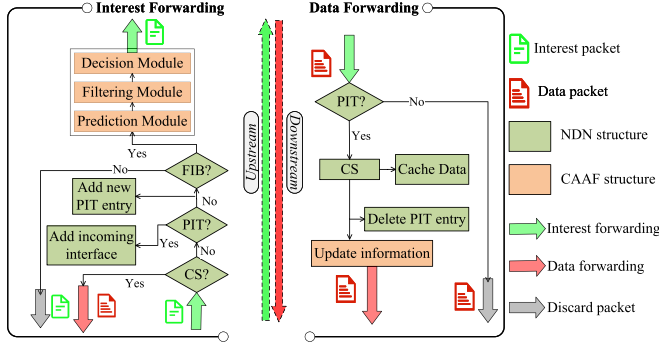


Fig. 3. Workflow of forwarding.

Step 5: Ranking and Deciding: Now the vehicles are ranked based on their similarity scores, with those closer to the positive ideal solution and further from the negative ideal solution receiving higher scores and ranks. Finally, the vehicle with the highest score is considered the best forwarder, i.e.,

$$F_i = \arg \max_{V_j \in \mathcal{N}(i)} S_j. \quad (31)$$

To show that this solution aligns with the theoretical optimal solution of (24), we analyze the relationship between them:

- The positive ideal solution is consistent with the maximum value of (24), while the negative solution is consistent with the minimum value of (24).
- A smaller D_j^+ indicates that V_j is closer to the positive ideal solution; a larger D_j^- indicates that V_j is farther from the negative ideal solution.

Therefore, maximizing S_j selects the node closest to the positive ideal solution (minimizing S_j^+) and farthest from the negative ideal solution (maximizing S_j^-), aligning with the objective of (24).

D. Workflow of Forwarding

The workflow of our proposed forwarding strategy is illustrated in Fig. 3. During the upstream, when an intermediate node receives an Interest packet, it first looks up the Content Store (CS) to find whether the requested content is already cached. If yes, the node returns a copy of the Data packet through the incoming interface and discards the Interest packet. Otherwise, the node performs an exact lookup match in the Pending Interest Table (PIT). If a matching entry is found, it append the incoming interface to the existing PIT entry and eventually discard the Interest packet. This serves to aggregate multiple Interest requests for the same content, as the node has already forwarded the same Interests which have not been satisfied yet. Otherwise, the Interest packet is forwarded based on our cache-aware adaptive forwarding strategy outlined in this paper. This process consists of three steps: predicting the probability of different vehicles in the network caching the requested content to identify content sources; filtering out the candidate set of eligible forwarding vehicles based on their geographical locations and motion information; and finally, selecting the optimal vehicle

Algorithm 1: Cache-Aware Adaptive Forwarding.

Input: Current vehicle V_i , requested content C_j , current time $t = nT + \tau$;

Output: The next-hop optimal forwarding node for retrieving content C_j : F_i ;

1 Cache Prediction:

2 for $V_j \in \mathcal{V}$ do

3 $k_j^m, \tau_j^m, \lambda_j^m \leftarrow \text{the } CIT_j(nT)$;

4 $P_j^m(t) \leftarrow (k_j^m, \tau_j^m, \lambda_j^m)$ according to Eq. (18);

5 end

6 Filtration of eligible forwarding vehicles:

7 Initialize the set $\mathcal{N}_i = \emptyset$;

8 for $V_j \in \mathcal{V}$ do

9 Calculate $L_{ij}(t)$ according to Eq. (19);

10 if $L_{ij}(t) > RTT_i(t)$ and $j \in \mathcal{R}_c$ then

11 $\mathcal{N}_i \leftarrow \mathcal{N}_i \cup V_j$;

12 end

13 end

14 Decision of the optimal forwarding vehicle:

15 for $V_j \in \mathcal{N}_i$ do

16 Calculate (d_j, l_j, r_j) ;

17 end

18 Construct normalized matrix $\mathbf{Z} = [z_{jk}]$;

19 $\vec{h}^+, \vec{h}^- \leftarrow \mathbf{Z}$ according to Eq. (28);

20 for $V_j \in \mathcal{N}_i$ do

21 $S_j \leftarrow \vec{h}^+, \vec{h}^-, \vec{h}_j$ according to Eq. (30);

22 end

23 $F_j \leftarrow S_j$ according to Eq. (31);

with the highest score based on multi-attribute decision-making within the candidate set.

During the downstream, when an intermediate node receives a Data packet, it examines the entry of the packet's name within its PIT to validate whether the node has previously forwarded an Interest packet for the specific content. If no match is found in the PIT, the node considers the data packet as unsolicited and discards it. On the contrary, if it discovers a matching entry in the PIT, the node multicasts the Data packet to all incoming interfaces recorded in the PIT entry, and deletes the corresponding PIT entry since the interest has now been satisfied. Subsequently, it updates the corresponding information including transmission delay and success ratio. Moreover, based on the caching strategy, the node also decides whether to cache a copy of the content.

E. Discussion

Computational Complexity: CAAF consists of three phases: cache prediction phase, filtering phase, and decision phase. In the cache prediction phase (Step 1 of Algorithm 1), for each vehicle in $\mathcal{V}$, a maximum of $|\mathcal{C}|$ additions and multiplications are required to compute the probability for each content item. Therefore, the computational complexity of the cache prediction is $\mathcal{O}(|\mathcal{V}| \cdot |\mathcal{C}|)$. In the filtering phase (Step 2 of Algorithm 1), it filters out the eligible forwarding vehicles from the entire set $\mathcal{V}$, which has a computational complexity of $\mathcal{O}(|\mathcal{V}|)$. The decision

phase (Step 3 of Algorithm 1) sorts the vehicles in the candidate set $\mathcal{N}(i)$ and selects the optimal vehicle. According to the multi-attribute decision process, the computational complexity of this step is $\mathcal{O}(|\mathcal{N}(i)|)$. Thus, the total computational complexity of Algorithm 1 is the sum of the complexity of steps 1 through 3, which is $\mathcal{O}(|\mathcal{V}| \cdot |\mathcal{C}|)$.

Communication Overhead: Regarding the communication overhead of the algorithm, we introduce a new data structure CIT for each vehicle. The structure of CIT is shown as Table II and the size of it is approximately 4B, which is much smaller than the Data packet. In addition, the time interval T for the CITs' synchronization is also much larger than the user's request rate. Consequently, the additional communication overhead incurred by CAAF is much smaller than that of the data transmission, and will not affect the data transmission.

The computational as well as communication overhead of the algorithm grows linearly with the number of vehicles. Due to the typically limited number of vehicles and content items, the additional overhead incurred by our proposed cache-aware adaptive forwarding strategy remains manageable and acceptable.

Implementation: Algorithm 1 presents the cache-aware adaptive forwarding procedure. It is straightforward to deploy in practical vehicular networks. First, the information exchange required for the Algorithm 1 consists of two main parts: exchange of movement information between vehicles and their neighbors; exchange of CITs between vehicles and RSUs. These exchanges can be handled using existing vehicular communication protocols, such as DSRC, without additional complex hardware or software support. Second, the three modules in Algorithm 1 (prediction module, filtering module, and decision module) can be integrated into the existing NDN forwarding engine without modifying the native NDN communication protocol. These modules can be installed alongside the NDN protocol stack on each vehicle node and automatically triggered during packet forwarding, enabling distributed deployment and operation. With this design, the implementation of Algorithm 1 is both scalable and practical, allowing seamless integration into existing vehicular network architectures.

V. PERFORMANCE EVALUATION

To demonstrate the advantages of our proposed CAAF strategy, we compare it with existing representative forwarding strategies in VANETs: VNDN [21], DASB [25], LISIC [26] and PRFS [28].

We implemented the comparison strategies and CAAF strategy in ndnSIM [38], which is an NS-3 based NDN simulator. For the sake of fairness, when comparing different forwarding strategies, we use the same mobility traces and network parameters, as shown in Table IV. We consider vehicular network topology in a 10 km two-way four-lane highway scenario and the mobility traces are generated through a Random Trip model by SUMO [39]. The communication range of each node is 200 meters. The communication technology adopts the IEEE 802.11p protocol, and the transmission rate is 6 Mbps. The consumer sends requests at a rate of ten packets per second and the content of the request obeys a Zipf distribution. The

TABLE IV
SIMULATION PARAMETERS

Parameters	Value
Simulation scene	Highway with 10 km
Max transmission range	200 m
Request rate	10 packets/s
Payload size of Data packet	1 kB
Transmission power	40 mW
Transmission bit rate	6 Mbps
Carrier frequency	5.9 GHz
Path loss model	Log distance path loss model
Communication technology	IEEE 802.11p
Cache replacement policy	LRU
Number of content	100
Simulation time	20 s

TABLE V
PREDICTION ACCURACY UNDER DIFFERENT PERIODS

Period (ms)	Accuracy	Interaction Numbers
500	0.9556	4000
1500	0.9508	1333
2500	0.9463	800
3500	0.9402	570
4500	0.9377	444

TABLE VI
PREDICTION ACCURACY UNDER DIFFERENT THRESHOLDS

Threshold P_{th}	Accuracy
0.4	0.9431
0.6	0.9497
0.8	0.9507
0.9	0.9474
0.95	0.8929

performance metrics utilized to evaluate the effectiveness of the forwarding strategy include the following:

- Forwarded Interest Packet (FIP): The average number of Interest packets forwarded by each vehicle.
- Forwarded Data Packet (FDP): The average number of Data packets forwarded by each vehicle.
- Interest Satisfaction (ISD): The average delay from the first time the consumer sends an Interest packet to the receipt of the corresponding Data packet.
- Interest Satisfaction Ratio (ISR): The ratio of satisfied Interest packets to the total number sent by consumers.

A. Prediction Accuracy Under Different Settings

Define $\eta = \frac{N_{true}}{N_{true} + N_{false}}$ as the prediction accuracy, where N_{true} is the number of correct predictions and N_{false} is the number of false predictions. By varying the values of CITs' distribution period T and the probability threshold P_{th} , we evaluate their impact on the accuracy of the prediction module, as illustrated in Tables V and VI. The other parameters are kept consistent throughout the experiments: requests sent by consumers followed a Zipf distribution with a parameter of 0.7, the number of vehicles in the network was set to 100, and each

vehicle's speed was randomly generated between 60 km/h and 80 km/h.

First, it can be observed that the accuracy of the prediction gradually decreases with the increase of the period. This trend arises because longer distribution periods result in less accurate CITs obtained by vehicles periodically, thereby reducing the accuracy of predictions derived from historical CITs. Conversely, as the period increases, the number of information interactions between vehicles and RSUs becomes less, resulting in a reduction in the additional overhead incurred. Therefore, a trade-off exists between prediction accuracy and overhead. Notably, even in the case of a long distribution interval (e.g., 4 s, which is 40x times the requesting rate), the prediction accuracy can remain high (above 90%), and even higher than 95% when the distribution period is 1 s. This shows that our prediction module can greatly improve the accuracy of the collected data and achieve excellent performance. Striking a balance between prediction accuracy and additional overhead, we set the distribution period $T = 2$ s in subsequent experiments.

Second, observations indicate that as the probability threshold grows, the accuracy of the prediction initially rises and then declines. This behavior can be attributed to the two types of prediction errors: False Positives, where a vehicle is incorrectly predicted to have cached content when it has not, and False Negatives, where a vehicle is incorrectly predicted to have not cached the content when it has. At lower threshold settings, raising the threshold initially reduces the number of False Positives, leading to an initial increase in overall prediction accuracy. Nevertheless, once beyond a certain optimal threshold, further increases in it trigger a detrimental rise in False Negatives, which in turn reverses the previously observed accuracy improvement. This phenomenon highlights the existence of a sweet spot in the probability threshold that optimally balances between minimizing both types of predictive errors. Based on the results of the experimental analysis, we set the probability threshold $P_{th} = 0.8$.

B. Performance Under Different Node Densities

First, by varying the numbers of nodes, we evaluate the performance of each forwarding strategy under different node densities, as illustrated in Fig. 4. In the simulation, two vehicles are randomly selected as the consumer and provider, and requests sent by the consumer follow a Zipf distribution with a parameter of 0.7. Furthermore, the speed of each vehicle is randomly generated between 60 km/h and 80 km/h.

The result presented in Fig. 4(a) demonstrates that CAAF significantly outperforms VNDN, DASB, and LISIC in terms of FIP, as well as superior to PRFS. The commonality between VNDN, DASB, and LISIC is that they all employ a broadcast listening approach to suppress redundant forwarding of Interest packets. Nevertheless, this passive approach is only capable of suppressing a limited amount of redundancy in the intersection region. In contrast, PRFS proactively selecting a single forwarding node separately for both forward and backward directions to markedly reduces overall redundancy, but still generates redundancy in the direction where no content provider

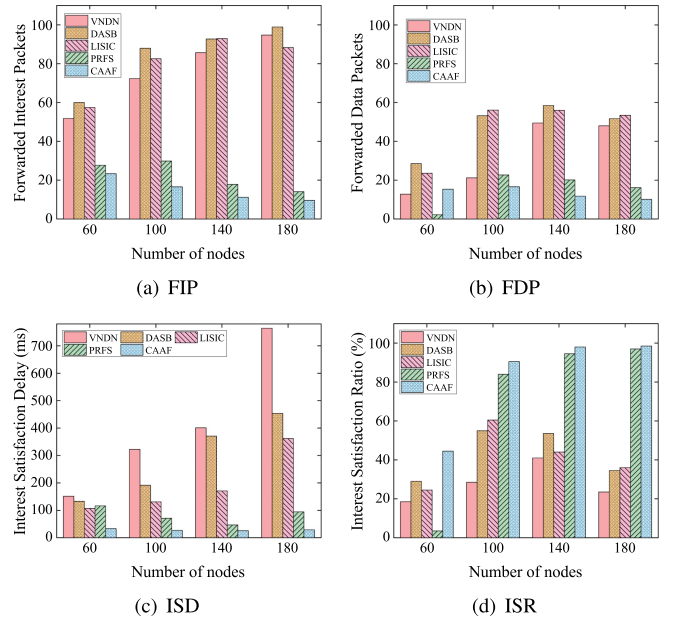


Fig. 4. Performance comparison versus different node densities.

exists. Furthermore, CAAF takes it a step further by allowing forwarding operations to be conducted only on the optimal node, thus minimizing redundancy to the greatest extent possible.

Similarly, it can be observed in Fig. 4(b) that the Data packet redundancy of CAAF is also much lower than that of VNDN, DASB and LISIC, as well as lower than that of PRFS. Notably, Fig. 4(b) highlights that under sparse densities (60 nodes), PRFS records lower FDP than CAAF. This phenomenon can be attributed to the fact that PRFS only successfully transmits a minimal number of Data packets at this point, which creates the “illusionary” of a lower FDP.

Moreover, with the increase in vehicle density, the number of neighbors of each vehicle increases, thereby leading to more redundancy for VNDN, DASB, and LISIC. Conversely, for PRFS and CAAF, the number of forwarding nodes selected is not affected by the number of neighbors, resulting in a more stable redundancy level with the change in vehicle density.

As illustrated in Fig. 4(c), the ISD of the VNDN, DASB, and LISIC strategies exhibits a gradual increase with the rise in node density. This is because the employment of a forwarding delay timer, which is inversely proportional to the distance. Consequently, as the node density increases, the distance between vehicles is shortened, resulting in a further increase in transmission delay. In contrast, PRFS and CAAF enhance the precision in selecting forwarding nodes as the node density increases, reaching the content source more rapidly. Moreover, CAAF considers the ISD factor in decision module, which allows for a further reduction in ISD.

In Fig. 4(d), we observe a shared trait among all strategies: a poor ISR at low node densities. This phenomenon can be attributed to the limited number of forwarding candidates and low link robustness under sparse node distribution. As the node density rises, the number of intermediate nodes grows, enhancing link stability and gradually improving ISR.

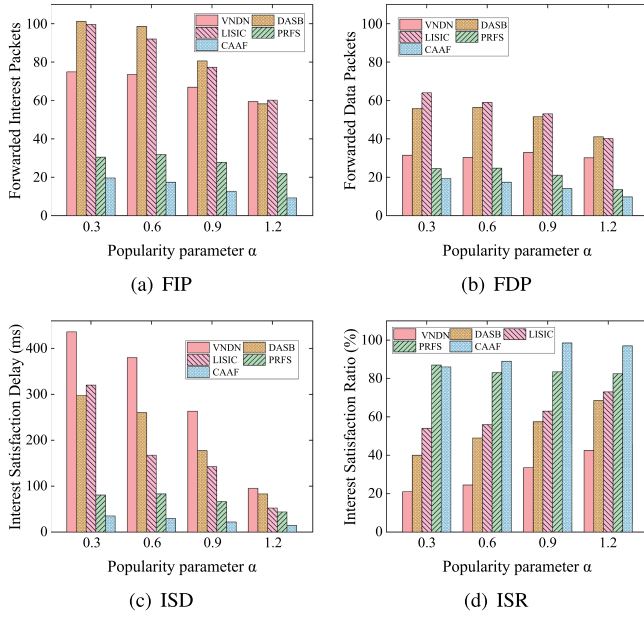


Fig. 5. Performance comparison versus different content popularity.

Nevertheless, at a too-high node density (180 nodes), VNDN, DASB, and LISIC experience a decline due to network congestion and collisions caused by excessive redundant traffic. On the other hand, PRFS shows a lower ISR under sparse conditions (60 nodes) since it executes forwarding via the most distant node, which introduces instability. In contrast, CAAF maintains consistently high performance across all node densities, with cache-aware capabilities and accurate forwarding decisions.

C. Performance Under Different Request Distribution

Next, by changing the parameter α of the Zipf distribution, we evaluate the performance of each forwarding strategy under different request distributions, as illustrated in Fig. 5. In the simulation, we use the same network topology and movement trajectory as in Section V-B when the number of nodes is 100.

Observations drawn from Fig. 5(a) and (b) reveal that with α increase, all strategies experience a reduction in FIP and FDP. This phenomenon is rooted in the concentration of consumer's requests on a few popular content as α increase, which in turn leads to these contents frequently being cached on vehicles within the network. Consequently, Interest packets for popular content are not required to be delivered to the far provider; rather, they can be satisfied by near cached copies within the network. This localization of content satisfaction reduces the number of node involvement in forwarding, thereby reducing redundancy. Furthermore, Fig. 5(c) and (d) illustrate a shared trend among all strategies: an enhancement in performance as the distribution parameter increases. Specifically, the ISD decreases due to the shortening of the forwarding path, while the ISR increases due to the gradual enrichment of the in-network cache.

Nevertheless, strategies such as VNDN, DASB, LISIC, and PRFS lack cache awareness, rendering them reliant on serendipity to encounter vehicles with cached copies during forwarding.

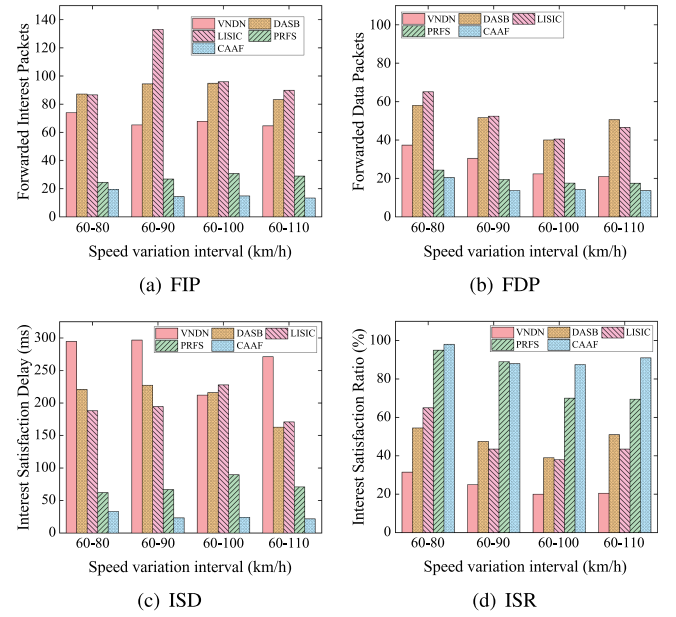


Fig. 6. Performance comparison versus different speed variations.

Conversely, our proposed strategy proposes a cached content prediction model, thereby endowing each vehicle with cache-awareness capabilities. Furthermore, we design an adaptive forwarding strategy that meticulously identifies the most advantageous forwarding node. Consequently, CAAF makes efficient use of in-network caching resources, resulting in significant improvements in content retrieval efficacy.

D. Performance Under Different Mobility Scenarios

Third, we evaluate the performance of each forwarding strategy under different mobility scenarios by changing the vehicle speed variations, as illustrated in Fig. 6. In the simulation, we use the same network topology as in Section V-C and the request follows a Zipf distribution with $\alpha = 0.7$.

As illustrated in Fig. 6(a) all strategies maintain a stable FIP when the vehicle speed variations vary since the vehicle speed does not influence the selection of the forwarding node during Interest packet transmission. Nonetheless, an increase in speed variations leads to a more volatile network topology with links becoming less robust, exacerbating the challenge of RPP. Consequently, the percentage of Data packets successfully transmitted decreases, as reflected in the trend of FDP in Fig. 6(b). Notably, given that vehicle velocities are randomly assigned, the resultant performance of individual strategies can display some oscillation. For instance, Fig. 6(a) illustrates a momentary rise in LISIC's FIP when the speed variation is 30 km/h, while Fig. 6(b) reveals similar fluctuations for both LISIC and DASB when the speed variation is 50, highlighting the sensitivity of these strategies to random changes in network conditions.

As shown in Fig. 6(c), the CAAF exhibits the smallest ISD for any speed variations, indicating its ability to provide faster data acquisition. VNDN introduces a forwarding delay timer, which results in an unnecessarily delay and the highest ISD. DASB

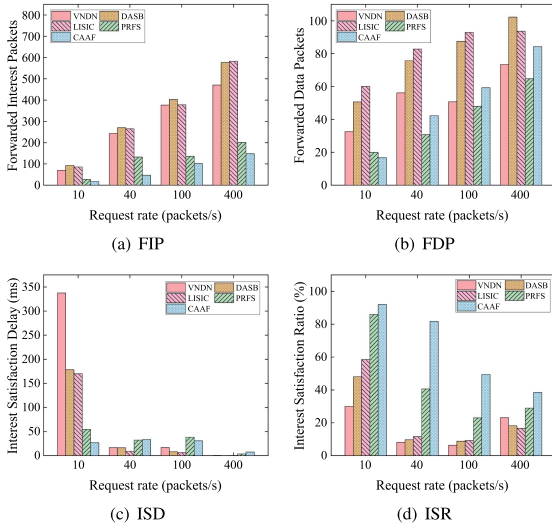


Fig. 7. Performance comparison versus different request rates.

lacks awareness of vehicle movement, which makes it difficult to select the correct forwarding node in mobile scenarios, leading to a high ISD. LISIC tends to select more stable nodes for forwarding, and therefore increases the number of transmission hops, which also results in a larger ISD. PRFS predicts vehicle movement through LSTM, which ensures a lower ISD in mobile scenarios. Furthermore, CAAF predicts the in-network caching, which makes the overall transmission path shorter and thus obtains the smallest ISD.

As illustrated in Fig. 6(d), the ISR of VNDN, DASB, LISIC, and PRFS all exhibit a declining trend as the speed variations increase. This is because as the speed differential between vehicles increases, the distance between individual vehicles changes more rapidly, leading to more unstable links and frequent changes in network topology. Lacking the capability to sense shifts in transmission conditions, these strategies struggle to accommodate the dynamics of an ever-changing network environment. In contrast, CAAF updates the network state of the corresponding node each time a packet is received, thereby dynamically sensing the network's evolving state. Moreover, it incorporates these metrics into its forwarding node selection process, empowering CAAF with a unique capacity to adapt to network changes, thereby maintaining superior ISR performance even at high vehicle speed variations.

E. Performance Under Different Request Rates

Finally, we evaluate the performance of each forwarding strategy under different request rates, as illustrated in Fig. 7. In the simulation, we use the same network topology and movement trajectory as in Section V-B when the number of nodes is 100 and the request of consumer follows a Zipf distribution with the parameter $\alpha = 0.7$.

As shown in Fig. 7(a) and (b), the FIP and FDP increase for all strategies as the request rate rises. This is because a higher request rate leads to an increase in the number of packets and also induces network congestion. Moreover, as depicted in Fig. 7(d), when the request rate is high, severe congestion results

in significant packet loss, causing the ISR to drop across all strategies. Under this case, CAAF consistently maintains the highest ISR due to its adaptive forwarding decisions, whereas other strategies struggle to ensure reliable transmission even at moderate rates (e.g., ISR falls below 20% at 100 packets/s).

It is noteworthy that, in Fig. 7(c), under high request rates, the ISD of strategies such as VNDN, DASB, and LISIC is slightly lower than that of CAAF. This occurs because the total number of content items is limited, and at high request rates, duplicate requests are more likely. These requested contents are often cached in nearby nodes, allowing flooding-based strategies to respond more quickly. However, due to their low reliability, this marginal ISD advantage does not translate into improved overall transmission performance.

VI. CONCLUSION

In this paper, we proposed an NDN-based cache-aware adaptive forwarding strategy for VANETs. Our strategy revolves around predicting cached content and ensures reliable content retrieval by adaptively selecting forwarding nodes that prioritize fast delivery and stable transmission. Specifically, we first predict the content cached in each vehicle, thereby providing necessary content source information for forwarding decisions. Second, a filter stage is executed to sieve candidate forwarding vehicles that meet both the accessibility and stability requirements. Finally, a multi-attribute decision-making method is employed to determine the optimal forwarding node. Our comprehensive simulation results highlight the strengths of the propose strategy, which consistently outperforms state-of-the-art forwarding strategies in VANETS across a variety of scenarios.

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